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Review

Implementation of clinical decision support tools for treatment selection in knee osteoarthritis: a scoping review

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***Abstract*:** Knee osteoarthritis (KOA) presents heterogeneous phenotypes, motivating a need for clinicians to deliver targeted therapies. There is a plethora of options that can be encompassed in KOA treatment regimes. Clinical decision support (CDS) tools that incorporate individual patient data have the capacity to tailor treatments to meet a patient's individual needs and assist with clinical decision-making. We aim to identify and evaluate CDS tools for individuals with KOA that use individualised prediction models to guide intervention decisions. A scoping review of the literature. A systematic search of six electronic databases, including Ovid Embase, Ovid Medline, Cochrane, CHINAL Ultimate, Scopus and Web of Science, was conducted for articles published between January 1, 2010 and May 17, 2024. Two reviewers independently screened articles and extracted data on study design, tool implementation and underlying prediction models. Eligible studies implemented personalised decision aids, designed to support clinical decisions regarding KOA interventions. The search yielded 5376 publications, of which 2445 were duplicates, leaving 2931 for screening. After title/abstract and full-text reviews, 14 studies were included in the final analysis, with one added through citation searching. Ten distinct decision aids were identified across the included studies. Most studies originated from the United States. Fewer than half of the decision aids included personalised information about non-surgical alternatives. Outcomes such as knee pain and physical function were the most commonly addressed, while psychosocial and financial impacts were rarely reported. Limited details were provided about the development and functionality of the underlying prediction models. Personalised decision aids for KOA show promise in supporting patient-centred decision-making. However, their clinical utility is constrained by limited transparency in model development and implementation. Future studies should emphasise the inclusion of non-surgical treatment options, early-stage KOA patients and personalised outcomes beyond pain and function to enhance their relevance and impact in clinical practice.

Keywords: clinical decision support tools, knee osteoarthritis, machine learning, scoping review, treatment outcome

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Background

Osteoarthritis (OA) is a chronic condition and a significant contributor to years lived with a disability amongst all musculoskeletal conditions.[1] In 2021, the global prevalence of OA was 606.9 million cases, with the knee being the most affected joint (66%), followed by the hand (22.2%) and other OA (7.8%).[2] This represents a 132.2%

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increase in total OA reported cases since 1990, with a forecast of a further 74.9% increase in knee osteoarthritis (KOA) by 2050.[3]

Historically referred to as a 'wear and tear' condition, OA is now widely recognised as a complex multifactorial disease involving the interplay between genetic, metabolic, biochemical and biomechanical factors.[4] Consequently, disease phenotypes present variably, resulting in heterogeneity in the level of joint dysfunction, pain, stiffness, functional limitations and loss of valued activities. A myriad of factors implicated in disease onset and progression have provided targets for developing therapeutic modalities. These include oral analgesics, intra-articular injections, nutritional supplements and the pleiotropic effects of other medications.[5] The current therapeutic strategies are not curative but are targeted at the amelioration of disease symptoms and attenuation of disease progression. For end-stage KOA, total knee arthroplasty (TKA) is considered an effective treatment,[6] with high success rates as evidenced by a range of objective medical outcomes including survival, prosthesis performance and revision rates.[7]

Over the last decade, a number of guidance documents have been produced by regional and international bodies that aim to assist clinicians in the management of OA cases.[8] In several guidelines, education, exercise and weight loss are indicated as appropriate 'first line' or 'core' treatments.[8,9] However, it has been noted that there has been a lack of implementation of such recommendations.[10,11] Despite the recent production of higher-quality guidelines globally, it has been noted that non-surgical OA disease management is delivered in an 'ad hoc way' that does not acknowledge variable disease phenotypes.[11]

There is an unmet need to support clinicians and patients to make targeted OA treatment decisions. In addition, a combination of treatments may be more suitable to enable a multi-targeted approach to effectively reduce pain and improve joint function than a single modality alone.[12] In musculoskeletal disorders, an increasing amount of patient data is being accumulated, including patient-reported outcome measures (PROMs). The increasing use of artificial intelligence (AI) and machine learning (ML) in clinical settings to interpret findings from large datasets has allowed users to learn from the data and deliver predictions in the form of clinical decision support (CDS) tools that can assist with clinical decision-making and treatment selection.[13] Incorporating individual patient data into such models provides the potential to tailor treatments to a patient's individual needs and provide meaningful treatment options.[14] Indeed, CDS tools can enhance shared decision-making (SDM) practices by providing treatment options to both clinicians and patients that can address individual patient needs. There is, however, a current gap in the literature of a consolidation of currently implemented CDS tools that focus on supporting treatment choices for OA, which can guide clinicians and provide insights regarding the clinical validity of tools that have been developed or are under development.

In the current scoping review, we synthesise the literature regarding implemented CDS tools for KOA that assist with treatment decision-making by incorporating patient data to enable treatment choices to be tailored to patients' individual needs. We chose the inductive and configurative scoping review approach versus systematic or mapping review methods to outline the breadth and characteristics of currently implemented CDS tools in KOA. This allowed us to explore the diversity of prediction models, implementation strategies and reported outcomes across a heterogeneous body of literature.[15] We anticipate that the findings from our review will guide clinicians and future research endeavours to address the need to help streamline therapeutic approaches for KOA patients, enhance SDM practices and potentially improve patient outcomes.

Methods

This scoping review was guided by the Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR).[16] The PRISMA-ScR Checklist was completed for this review and is available in Supplemental Appendix A. While this review systematically catalogues implemented CDS tools, a function often associated with mapping reviews, the iterative nature of our analysis, detailed data extraction and aim to identify inductive knowledge gaps aligns with scoping review methodology, outlined by PRISMA-ScR and described in a recent commentary.[15]

Eligibility criteria

Articles published from January 2010 until the search was executed on May 17, 2024, were included in this review, with the selected search

dates reflective of the upsurge in AI/ML.[17] Only peer-reviewed articles were included to align with our aim of analysing the different measures of implementation success.
The main inclusion criteria for articles were:

(1) If they considered individuals with KOA.
(2) Used a CDS tool/patient decision aid (PtDA) to predict outcomes.
(3) Incorporated these predictions into decisions regarding the intervention/treatment strategy.

Papers were excluded if:

(1) They considered rheumatoid arthritis, inflammatory arthritis or any arthritis not related to the knee, high-risk individuals (i.e. no current KOA diagnosis), revision surgeries or knee surgery relating to other soft tissue pathologies (i.e. anterior cruciate ligament replacement).
(2) There was no prediction involved, including if preferences were the only patient-based information included or if general guides were used (i.e. matching to similar patients based on only a single patient dimension).
(3) Articles were not peer-reviewed, were review articles, conference presentations, abstracts, protocols, opinion or perspective pieces.
(4) Articles were not available in English.

Information sources and search strategy

The search strategy and sources of information were decided upon by the research team in consultation with a member of the library services team, University of Newcastle, Australia.

Six electronic databases were searched, including Ovid Embase, Ovid Medline, Cochrane, CHINAL Ultimate, Scopus and Web of Science. The search terms related to three main concepts:

(1) KOA or TKA patients.
(2) CDS tool/PtDA.
(3) Patient/personalised outcomes or intervention strategies.

The final search strategy was developed by two authors (J.A.C. and O.R.) and reviewed and agreed upon by the research team. A full description of the electronic search strategy is presented in Supplemental Appendix B. The main search concepts reflect our narrower focus on a broad question, aligning with a scoping review methodology. Citation/reference list searching and forward searching of included articles were iteratively conducted to identify other relevant publications not found through the database search. Extracted articles were imported into Clarivate EndNote 20.,[18] and the study screening process was conducted via Covidence.[19]

Selection of sources of evidence

Two authors (J.A.C. and O.R.) independently conducted the screening process. Duplicate articles were first removed from the results of the database search, which were then screened for title and abstract based on the inclusion/exclusion criteria. Discrepancies were resolved by consensus or by a third author (K.R.) if needed. Articles that were included from the title and abstract screening process were then reviewed in full based on the inclusion/exclusion criteria.

Data extraction and data items

Two authors (J.A.C. and O.R.) conducted the detailed data extraction independently, with any discrepancies resolved through a discussion. Key information extracted from articles was based on three main domains: (1) Study design, including type of CDS tool, expected user, mode of delivery, setting, study type, population; (2) Predictive modelling, covering the dataset/registry used, number of patients/datapoints, type of prediction model and statistical methodology, patient inputs and outputs, timepoints and model performance; and (3) Implementation, sample size, measured outcomes, control conditions, treatment options presented, end user of the tool and when it is completed, time taken to complete the tool, implementation timepoints and clinical success. The detailed data extraction process, based on pre-defined charting fields, was designed to support a configurative synthesis of the heterogeneous evidence in this field.

Synthesis of results

Extracted data were presented in tabular form with further narrative synthesis based on the study characteristics. Three main themes were used to discuss and compare the studies:

(1) The study design and demographic characteristics.
(2) Aspects regarding the development and validation of the underlying prediction model.
(3) The success and attributes of the clinical implementation and selected outcome measures.

Results

Selection of included studies

In total, 5376 articles were extracted. After the removal of duplicates ($N=2445$), 2931 publications underwent screening for inclusion. Both the title and abstract of each publication were screened based on the inclusion/exclusion criteria, resulting in 2873 articles being excluded. Full-text screening of the remaining 58 articles was then undertaken independently by two authors (J.A.C. and O.R.) and resulted in a further 45 articles being excluded, leaving 13 articles. An additional 30 articles were identified through forward and backward citation searching. Publications were retrieved and assessed for eligibility by two authors (J.A.C. and O.R.). Of those identified, one additional article was included, leaving a total of 14 articles. A summary of the search and screening process is presented in Figure 1.

Characteristics of included studies

The 14 included studies covered 10 different decision tools, each with various characteristics. Table 1 provides information on the characteristics of each study. Tables 2 and 3 describe the characteristics of the interventions and prediction models considered respectively.

Study design and location. The included studies considered KOA patients across a diverse geographical range with varying demographic features (Table 1). Seven of the included studies were conducted on a US cohort,[20–26] four studies using an Australian cohort,[27–30] and two studies a Canadian cohort. Two of the studies conducted in Australia allowed online recruiting, such that participants were from a number of countries.[28,30] One study did not clearly report the demographic characteristics of its participants,[31] and none of the included studies were conducted in an Asian-centric population.

While all studies included KOA patients, the targeted disease stage varied. Most interventions were based on KOA patients at the stage of considering TKA or awaiting TKA. Three decision aids were considered for KOA patients deciding on different treatment options, but not necessarily at the end-stage point.[21,23–26,31] For one PtDA, patients were included if they had received a diagnosis of KOA but had not yet visited their practitioner to discuss possible treatment options.[28,30]

Five studies employed a randomised control trial (RCT) to investigate the effect of the intervention versus a control comparator group.[20,21,29,32,33] The nature of the control group varied between studies and comprised routine care, treatment as usual or no PtDA or education or general information. The remaining studies were either qualitative studies[23–26,30,31] or cohort studies.[22,27,28]

Decision aid features. The mode of delivery for PtDAs was similar across the studies, with most implementations completed by the patient via a web-based interface or tablet application[23–33] (Table 1). Other variations to access included via group meetings,[22] a software platform at the clinic prior to consultation,[20] or a clinical study coordinator at consultation, with a printout of the results provided to the patient for further discussion.[21]

The end user of the decision aid also varied across the studies (Table 1). The majority of studies were designed to provide information to the pati ent,[21–26,28–31] three to both the patient and surgeon and one to the surgeon only. In some studies, predicted outcome measures were presented alongside additional information. Four studies presented education or information about OA, with preference information also presented by Jayakumar et al.[20] with the intention for the tool to motivate SDM.

Treatment options. The presented treatment options also differed between the decision aids, as shown in Table 2. All decision aids presented TKA as a personalised treatment option. For three PtDAs, the only treatment presented was surgery.[23,27,29] Of those that presented non-surgical options, most presented generic, non-personalised information about alternative treatments.[20,22,24–26,32,33] The options presented were specified for one decision aid (physical therapy, medications, injections)[24–26]; however, further details relating to non-surgical interventions were not provided for other decision aids.[31] Only

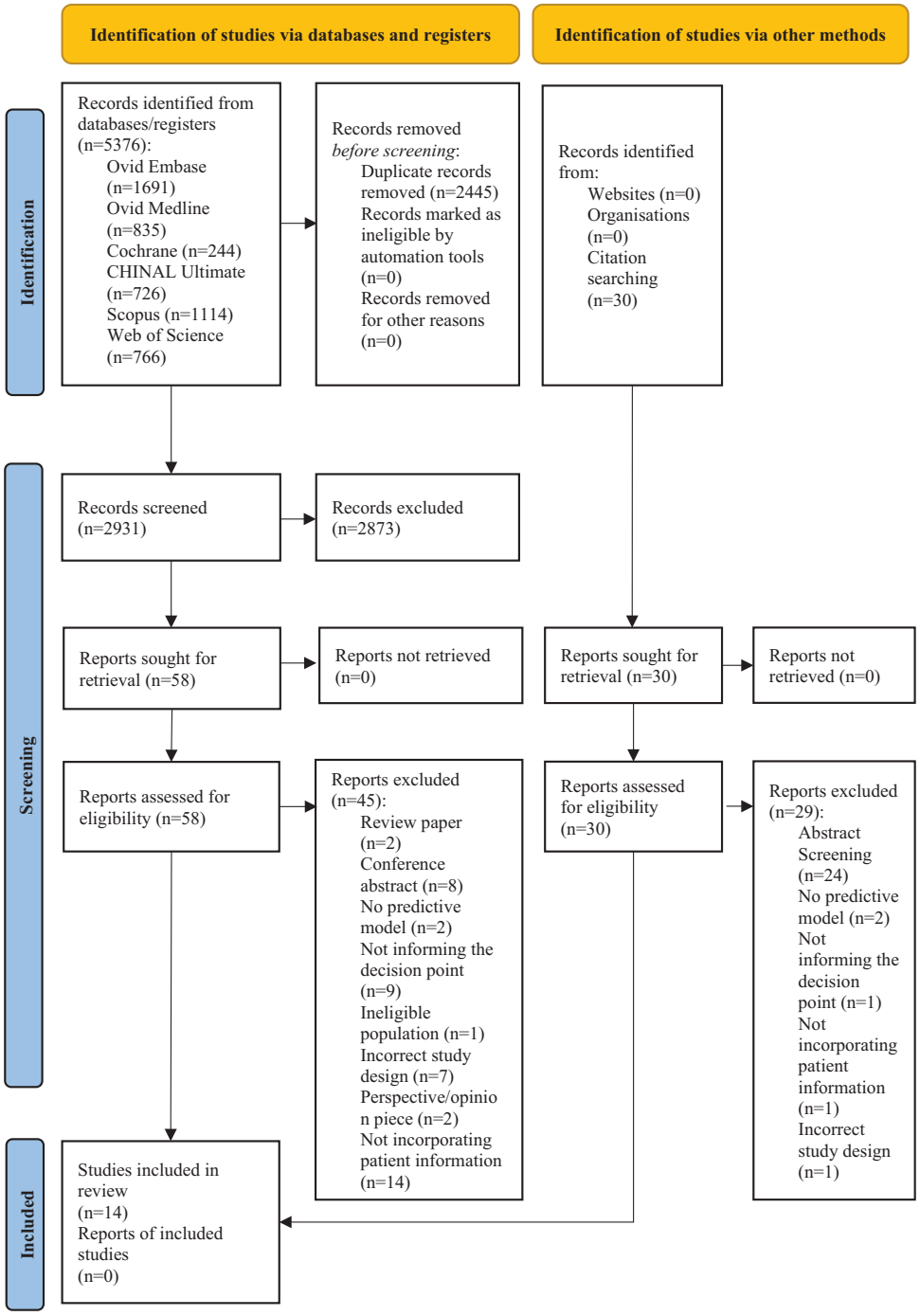


Figure 1. PRISMA flow chart[16] describing article selection and inclusion process.
PRISMA, Preferred Reporting Items for Systematic reviews and Meta-Analyses.

two PtDAs provided personalised information about alternative treatments. One provided up to 20 different options, including medication, nonsteroidal anti-inflammatory drugs (NSAIDs), injections, exercise, education and weight loss, depending on the patient's choices.[28,30] The other presented three options, two of which were selected by the clinician from either weight loss, exercise, medications, injections, NSAIDs, knee brace or TKA, with no treatment as the third option.[21]

Outcome measures. Outcomes from both the prediction model and the intervention were measured by the included studies. There was little

Table 1. Characteristics of included studies.

Study reference	Study type	Patient group	Setting/ population	Mode of delivery	End user	Name of CDS tool
Bansback et al. (2022)[32]	RCT	TKA patients	Canada	Online	Patients and surgeons	–
Marshall et al. (2023)[33]	RCT	KOA considering TKA	Canada	Online	Patients and surgeons	–
Jayakumar et al. (2021)[20]	RCT	Patients with OA-related knee pain	United States	Software platform	Patients and surgeons	–
Twiggs et al. (2019)[27]	Cohort study	TKA patients	Australia	Tablet app/web application	Surgeon	–
Moreton et al. (2022)[28]	Cohort study[a]	OA patients	Australia	Web-based	Patients	MyJointPain (via Annalisa)
Jeon et al. (2019)[30]	Qualitative	OA patients	Australia	Web-based	Patients	MyJointPain (via Annalisa)
Jones et al. (2022)[21]	RCT	KOA patients	United States	Printed report	Clinical study coordinator and patients	–
Zhou et al. (2024)[29]	RCT	KOA patients awaiting TKA	Australia	Online	Patients	SMART Choice Knee Tool
Volkmann and Fitzgerald (2015)[22]	Cohort study[b]	Severe KOA patients	United States	Group meeting	Patients	–
Daudelin et al. (2020)[31]	Qualitative	Severe KOA patients	Not specified	Web-based	Patients	KOMET
Zheng et al. (2018)[23]	Qualitative	KOA patients considering treatment	United States	Web-based	Patients	–
Franklin et al (2021)[24]	Qualitative	KOA patients considering treatment	United States	Web-based + printed report	Patients	A.S.K
Stern et al. (2022)[25]	Qualitative	OA patients	United States	Web-based + printed report	Patients	A.S.K
Pila et al. (2023)[26]	Qualitative	KOA patients	United States	Web-based + printed report	Patients	A.S.K

[a]Information not specified in paper; study designed such that decision aid is available online and accessible for anyone with the link, although distributed to OA groups.
[b]Information not specified in paper; study designed such that patients completed questionnaires at time of screening and invited to participate in a group meeting to receive the intervention.
CDS, clinical decision support; KOA, knee osteoarthritis; OA, osteoarthritis; RCT, randomised control trial; SMART, St Vincent's Hospital Arthroplasty Outcomes; TKA, total knee arthroplasty.

consistency across the studies regarding the measures of intervention outcomes, but a general divide was observed between those that assessed decision-making and those that collected PROMs (Table 2). Decision-making and experience metrics were considered by nearly all studies; the most measured outcomes were decision quality, as measured by the Knee Osteoarthritis Decision Quality Instrument[20,29,32] or via an interview,[24] and decisional conflict, as measured by the Decisional Conflict Scale[22,25] or the SURE (Sure of myself, Understand information, Risk-benefit ratio, Encouragement) test.[32] Other measures included willingness for treatment,[32] level of SDM (CollaboRATE survey),[20,32] and patient recall.[22,24,25] The five RCTs assessed decision-making or experience-based metrics as their primary outcomes.[20,21,29,32,33]

Table 2. Summary of personalisation of interventions.

Related studies	Patient numbers	Treatment options presented	Information presented	Time points	Measured outcomes	Success of CDS implementation
Bansback et al. (2022)[32] Marshall et al. (2023)[33]	80 (Control – routine care), 75 (intervention)[32] 61 (Control – routine care), 71 (intervention)[33]	Personalised: TKA Non-personalised: non-surgical treatment	Information about OA and treatment options. Details on similar patients who had TKA. One page surgeon summary report on patient's knowledge and preferences.[32] Individualised PROMs, one page summary report (to surgeon).[33]	Not specified[32] 1 Year[33]	Decision-making: Modified HK-DQI, 4-item SURE Test, 3-item CollaboRATE, involvement preference (CPS), willingness for surgery[32] Decision-making: Patient expectations, 5-item DRS Satisfaction: Patient satisfaction Global Health: WOMAC, EQ-5D-5L Mental Health: PHQ-9[33]	Intervention arm had better odds of making a quality decision, being knowledgeable, less decisional conflict on whether to have surgery. Fewer were undecided about treatment. No significant difference in expectations being met, decisional regret or being satisfied. EQ-5D-5L significantly different between those who underwent surgery and those who did not.
Jayakumar et al. (2021)[20]	69 (Control – education only), 60 (intervention)	Personalised: TKA Non-personalised: non-surgical treatment	Education, preferences and personalised outcomes	Pre-consultation, post-consultation, 4–6 months	Decision-making: K-DQI (Q 3.1–3.5), CollaboRATE survey, consultation satisfaction (NRS), consultation time, TKA rate, treatment concordance K-DQI (Q 1.6) Physical Health: KOOS-JR	Intervention arm reported better decisional quality, collaborative decision-making, satisfaction and functional outcomes. No significant difference in consultation times, TKA rates or treatment concordance.
Twiggs et al. (2019)[27]	75 (Control – surgeon's standard practice), 75 intervention)	Personalised: TKA	Preoperative and postoperative KOOS pain score, a set of boxes detailing positive and negative prediction factors	12 Months	Physical Health: Change in patient-reported pain between those booked and not booked for surgery, accuracy of predictions	No significant changes to proportion of patients booked for TKA surgery. Statistically significant change in level of patient-reported pain between booked and non-booked. At-risk patients found to be 19.3 times more likely to not meet the MCID.
Jones et al. (2022)[21]	52 (Control – general information), 52 (intervention)	Personalised: No TKA and two of either weight loss, exercise, medications, injections, NSAIDs, knee brace, TKA	General information, printed tool output	Post-intervention, 1 month	Experience: PES, PIS, likelihood of recommendation, assessment of SDM tool Global Health: Specific behaviour changes Physical Health: Activity level	No significant difference in overall experience, likelihood of recommendation or taking recommended medication. Activity level was significantly higher in intervention group than control group.

(Continued)

Table 2. (Continued)

Related studies	Patient numbers	Treatment options presented	Information presented	Time points	Measured outcomes	Success of CDS implementation
Moreton et al. (2022)[28] Jeon et al. (2019)[30]	705 (Intervention, 625 short form, 180 long-form)[28] 36[30]	Personalised: Surgery, medication, NSAIDs injections, exercise, education, weight loss[28] Personalised: Multiple options from exercise to surgery[30]	Scores for each options presented graphically	Post-interventio	Not specified[28] Experience: Relative advantage, compatibility, complexity, trialability, observability, outcomes of innovation, adoption[30]	Patients who used aid reported a positive experience (64%), the tool changed their mind (38%), and use would improve the quality of decision-making (48%) Participants saw practical benefits, such as new insights and monitoring progress.
Zhou et al. (2024)[29]	106 (Control – treatment as usual), 105 (intervention)	Personalised: TKA	Predicted outcome report, probability score (0%–100% likelihood of improvement)	Baseline, post-intervention, 6 weeks, 12 weeks and 6 months	Decision-making: Willingness for surgery, treatment preference and K-DQI	No significant difference in willingness for surgery or change in decision quality markers.
Volkmann and Fitzgerald (2015)[22]	111 (Intervention)	Personalised: TKA Non-personalised: Surgical and non-surgical options	45-Min videotape and personalisedarthritis report	Post-intervention	Decision-making: Decisional readiness (6-point Likert scale), 16-question DCS, patient recall Expectations: adapted version of WOMAC	Improved physical function scores (non-significant), decision readiness, decisional conflict, recall and pain expectation.
Daudelin et al. (2020)[31]	14 (Development), 15 (final version)	Personalised: TKA, non-surgical options	Current and predicted values (with uncertainty) for outcomes with explanation	Post-intervention	Experience: Usability, engagement	86% Remained engaged throughout the project. 80% (60%) understood values on bar graphs (circles), 50% (50%) understood uncertainty.
Zheng et al. (2018)[23]	28 (Round 1: 8, round 2: 20)	Personalised: TKA	Metrics meaningful to patients, such as pain relief at rest and when walking, ability to walk five blocks 1-year post-op, home and community activity levels	While using or post-intervention	Experience: Design ideas, clarity of information, usefulness, satisfaction, acceptance	Highest satisfaction with the use of the tool and lowest with error messages. 89% of patients would recommend to a friend.

(Continued)

Table 2. (Continued)

Related studies	Patient numbers	Treatment options presented	Information presented	Time points	Measured outcomes	Success of CDS implementation
Franklin et al. (2021)[24] Stern et al. (2022)[25] Pila et al. (2023)[26]	3157 (Ongoing)[24] 25[25] 25[26]	Personalised: TKA Non-personalised: physical therapy, medications, injections	Patient's current scores compared to national sample and risk factors. Predicted postoperative pain, function, general health, inpatient care required. Generic decision grid of non-operative options	Post-intervention[24] 6 and 12 months[25] Within 6 weeks[26]	Decision-making: Recall, value of information, report use and decision quality[24] Physical Health: Non-specified PROMs assessments[24] Decision-making: Decision, DCS, recall[25] Experience: Content, presentation of data, engagement[26]	68% Recalled reviewing report 2 weeks post, majority of patients valued the information Patients found benefits in providing information, fostering communication and building confidence and trust Report satisfactory to most participants.

To avoid repeating text, overlapping colours resemble two or more studies that concern the same CDS tool.
CDS, clinical decision support; CPS, Control Preference Scale; DCS, Decisional Conflict Scale; DRS, Decisional Regret Scale; EQ-5D-5L, European Quality-of-Life 5-Dimensions 5-Levels; HK-DQI, Hip/Knee Osteoarthritis Decision Quality Instrument; K-DQI, Knee Osteoarthritis Decision Quality Instrument; KOOS-JR, Knee Injury and Osteoarthritis Outcome Score for Joint Replacement; MCID, minimal clinically important difference; NRS, Numerical Rating Scale; NSAIDs, nonsteroidal anti-inflammatory drugs; OA, osteoarthritis; PES, Patient Experience Survey; PHQ-9, Patient Health Questionnaire-9; PIS, Patient Information Survey; PROMs, patient-reported outcome measures; SDM, shared decision-making; SURE, Sure of myself, Understand information, Risk-benefit ratio, Encouragement; TKA, total knee arthroplasty; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index.

Table 3. Information regarding the model/method used to generate patient outcome information to inform the decision-making point.

Study reference(s)	Prediction model/method	Predicted patient outcome(s)	Patient input information	Dataset/ registry used	Patient numbers	Time points	Model performance
Bansback et al. (2022),[32] Marshall et al. (2023)[33]	Matching condition (based on input thresholds)[a]	Global Health: Change in EQ-5D	Demographics: Age, sex Clinical Factors: BMI Global Health: EQ-5D, WOMAC Mental Health: PHQ-9	Previous RCT data,[33] Alberta Bone & Joint Health database	Not specified	6 Months	Not specified
Jayakumar et al. (2021)[20]	Not specified	Global Health: Short-term complications Physical Health: Function, joint pain, stiffness, KOOS-JR	Demographics: Age, sex Clinical Factors: BMI, smoking status, comorbidities, no. ED visits, no. hospitalisations Global Health: PROMIS Global-10 Physical Health: KOOS-JR	Not specified	Not specified	7/30/90 Days (complications), others not specified	Not specified
Twiggs et al.1 (2019)[27]	Tree-augmented Naïve Bayes	Physical Health: Knee pain score	Demographics: Age, gender Physical Health: KOOS (ADLS, pain and symptom scores), pain (pivoting, standing, bending knee), difficulty standing, difficulty bending knee fully, back pain (frequency and severity), hip pain, falls in preceding year	OAI	330	12 Months	Development: $R^2 = 0.23$ Validation: $R^2 = 0.29$, $\rho = 0.53$
Moreton et al. (2022),[28] Jeon et al. (2019)[30]	Weighted-sum approach	Global Health: Side effects Physical Health: Pain, function Societal Costs: Costs to patient	Includes the following: Demographics: Age Clinical Factors: Joints most affected, comorbidities Preferences/expectations: Patient preferences	Not specified	Not specified	Not specified	Not specified
Jones et al. (2022)[21]	Markov model	Societal Costs: Financial impact of decreased productivity Physical Health: Knee pain, activity level	Includes the following: Demographics: Age, race, ethnicity, gender, education, insurance carrier Clinical Factors: Selected comorbidities (i.e. hypertension, obesity, diabetes) Physical Health: Pain, function	Multiple databases	Not specified	1, 5 and 10 years	Not specified
Zhou et al. (2024)[29]	Logistic regression	General: Likelihood for improvement	Demographics: Age, gender, ethnicity, socioeconomic indicator, remoteness Clinical Factors: BMI, indication for TKA, comorbidities (CCI)[b]	SMART Registry	3703[b]	12 Months[b]	AUC 0.712, accuracy 0.698, sensitivity 0.768, specificity 0.575, PPV 0.758, NPV 0.589

(Continued)

Table 3. (Continued)

Study reference(s)	Prediction model/method	Predicted patient outcome(s)	Patient input information	Dataset/ registry used	Patient numbers	Time points	Model performance
Volkmann and Fitzgerald (2015)[22]	Regression (mixed linear effects)[c]	Physical Health: Pain and function scores (SF-36 pain and physical function scores[c])	Demographics: Age, sex Physical Health: Pain and function scores	Dataset from Fitzgeral'd's 2004 study[c]	81[c]	1, 6, 12 Months	Not specified
Daudelin et al. (2020)[31]	Multivariable linear regression models	Physical Health: Knee pain (WOMAC knee pain scale), overall function (SF-12 PCS)	Demographics: Age, gender Clinical Factors: BMI, comorbidities (CCI $\geqslant$ 1), treatment (TKA or no TKA) Mental Health: SF-12 MCS Physical Health: SF-12 PCS, WOMAC knee pain, knee pain (contralateral), hip pain or pain/ ache/stiff	MOST, OAI, NEBH Registry and TMC Registry[d]	1452[d]	1 Year	Adjusted $R^2 = 0.32$ (WOMAC model), adjusted $R^2 = 0.34$ (SF-12 PCS model)[d]
Zheng et al. (2018)[23]	Not specified	Physical Health: Improvement in pain relief and physical function	20 Questions related to: Demographics Global health Physical Health: Knee pain and function Clinical Factors: Medical comorbidities Preferences/ expectations: Expectations one year after surgery	Not specified	Not specified	Not specified	Not specified
Franklin et al. (2021), Stern et al, (2022), Pila et al. (2023)[23–26]	Multivariable models	Physical Health: Pain, physical function, general physical health	Demographics: Age Clinical Factors: BMI, comorbidities (including diabetes), narcotics use, smoking history Mental Health: emotional health Physical Health: lower back pain[e]	FORCE-TJR national registry	Not specified	12 Months	Not specified

[a]Information found in Bansback et al.[34]
[b]Information found in Zhou et al.[35]
[c]Information found in Weng et al.[36]
[d]Information found in Selker et al.[37]
[e]Additional inputs mentioned in Franklin et al.[24] (knee pain at rest and walking, history of ipsilateral knee procedure), however unclear if used in final model.
ADLS, Activity of Daily Living Scale; AUC, Area under Curve; BMI, body mass index; CCI, Charlson Comorbidity Index; ED, emergency department; EQ-5D, European Quality-of-Life 5-Dimensions; FORCE-TJR, Function and Outcomes Research for Comparative Effectiveness in Total Joint Replacement; KOOS, Knee Injury and Osteoarthritis Outcome Score; MCS, Mental Component Score; MOST, Multicenter Osteoarthritis Study; NEBH, New England Baptist Hospital; NPV, negative predictive value; OAI, osteoarthritis initiative; PCS, Physical Component Score; PHQ-9, Patient Health Questionnaire-9; PPV, positive predictive value; PROMIS, Patient-Reported Outcomes Measurement Information System; RCT, randomised control trial; SF-12, Short Form 12-item Survey; SMART, St Vincent's Hospital Arthroplasty Outcomes; TKA, total knee arthroplasty; TMC, Tufts Medical Center; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index.

Interestingly, there were mixed results relating to the implementation success across the included studies, even when measuring the same outcome (Table 2). A secondary analysis of the PtDA presented by Bansback et al.[32] reported no statistical benefit of the decision aid implementation,[33] whereas the intervention arm in the primary analysis showed improved decision quality.[32] Similarly, Zhou et al.[29] reported no significant reduction in patient's willingness for surgery or change in decision quality markers, although other studies showed a positive impact on decision quality.[20,22] All qualitative-based studies reported positive feedback from the implementation[23–26,30,31]; however, in two studies, no significant differences were seen between the intervention and control arms in patient experience and expectation metrics.[21,33]

A range of PROMs were predicted as part of the information presented to the end user (Table 3). Pain, either an absolute value or an improvement, was the most common,[20–28,31] followed by activity level or function.[20–26,28,31] Other outcome measures included global health measures,[33] non-specific PROMs,[24] and financial impacts.[21,28,30] One study presented a general likelihood of improvement if the patient was to undergo TKA.[29]

Predictive modelling. The variables used as inputs for the prediction model varied between included studies but mostly captured the same domains (Table 3). All studies included demographics as inputs to the prediction model. This was predominantly age and sex, but some studies also included ethnicity,[21,29] education and insurance information[21] or socioeconomic indicators and remoteness.[29] Clinical factors were also used as inputs, with the most common being body mass index,[20,24–26,29,31–33] comorbidities,[20,21,23–25,26,28–31] and smoking status.[20,24–26] Most models included physical health measures, including Knee Injury and Osteoarthritis Outcome Score (KOOS)/KOOS-JR,[20,27] Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) pain,[31] or pain[21–26,31] and function[21–23,31] questions. Global health was also an input to some prediction models, captured by European Quality-of-Life 5-Dimensions and WOMAC,[32,33] Patient-Reported Outcomes Measurement Information System (PROMIS) Global-10,[20] or was unspecified.[23] Lastly, some studies also used mental health[24–26,31–33] and preferences/expectations[23,28,30] as model inputs.

Most studies either reported the method or model used to generate the predicted patient outcomes or gave a reference to a previous publication (Table 3). Although one study mentioned using an AI model to generate outcome predictions, no specific details or further references were provided.[20] The most common method for generating individualised predictions was via either linear or logistic regression.[22,24–26,29,31] Other approaches include Markov models,[21] a multi-dimensional matching condition based on similar patients in the dataset,[32,33] and a weighted-sum approach based on data from a meta-analysis.[28] One study took a distinctly nonlinear approach using the tree-augmented naïve Bayes model and included both the development and clinical validation of the predictive tool in the same article.[27]

Discussion

With diverse phenotypes seen across the KOA disease course, and the vast array of KOA non-surgical treatment options currently available, CDS tools have the potential to enhance SDM regarding treatment options and support the tailoring of disease management strategies to individual patient needs. In this scoping review, we identified 14 peer-reviewed articles that describe the features of CDS tools at the implementation stage that have used patient-based predictive modelling to inform KOA treatment strategies. We explored the potential clinical utility of the CDS tools by qualitatively examining their reported implementation characteristics. We also assessed the transparency and appropriateness of the underlying predictive modelling methodology, as well as the level of validation and prediction accuracy in the implementation phases of the tool development. Our synthesis revealed that there are significant limitations in both the predictive modelling and implementation stages across most studies identified, which limit the clinical utility of these CDS tools in guiding treatment choices. The limitations we have found provide insights for future CDS tool development as well as informing clinicians of the areas that should be addressed when considering the potential use of these tools in the clinic setting.

Clinical utility of decision aids

An important consideration for any implemented CDS tool is the possible impact on human

interaction between the patient and clinician.[38] One way to formalise this is through measuring any observed changes in time during the consultation period, although only one study considered this.[20] This quantification would help make comparisons between different clinical implementations. Alternatively, some studies[22,25,26,29] have been designed such that information is provided to the patient prior to consultation, easing the time burden during consultation. Ease-of-use for patients could also be enhanced via the collection and processing of data prior to consultation, a growing possibility with the emergence of trained AI chatbots and virtual agents.

Another improvement to ease-of-use could be obtained by addressing the number of predictors required for the CDS tool. Some studies noted that a subset of questions from those originally considered were important in predicting relevant outcomes.[27,29] Reducing the number of questions to be completed by the patients could improve the time required for data collection and may be particularly important for those decision tools that are completed with the clinician. Further benefits include reducing possible patient test fatigue and improving the likelihood of clinical uptake.

Treatment options

Many studies exist that predict outcomes for patients after TKA. However, few studies look at methods to incorporate predictions into the decision-making process for non-surgical treatment alternatives. Alternatives are often provided as generic information rather than being prediction-based, a criticism from participants in a qualitative study.[26] Considering the reported success of conservative measures earlier in the KOA trajectory,[39] it is important that these alternatives are considered in the development of future CDS tools. Moreover, it has been suggested that patients may be more receptive to CDS tools earlier in the disease trajectory,[32,33] and intervening at an earlier point in the KOA progression may have benefits for both time and costs for healthcare centres.[40,41]

Measures of intervention success

A wide range of metrics were used across the included studies to quantify the success of the clinical implementation. The outcome of interest depends on the objectives of the study and the intended use of the CDS tool. The included RCTs primarily looked at whether a significant change in a decision-making outcome measure was experienced depending on whether the patient was in the intervention or control group.[20,21,29,32,33] These outcomes implicate the clinician and stakeholders more than the patient. This, together with feedback on the presentation, could be used to justify the implementation of the given CDS tool. In addition, domains such as global health and functional outcomes were considered as outcomes by multiple studies.[20–22,24,27,33]

Transparency of reporting

Few studies clearly reported the underlying prediction model and retained input features, an impactful omission when interpreting the implementation phase of a CDS tool. As reported in Table 3, this information may have been included in the original article or in an existing publication. However, it is concerning that not all studies provide explicit details regarding model specifications. Furthermore, it is important to be aware of possible complexities and limitations (i.e. linear assumptions, dataset quality) of the implemented model. For example, prediction models such as linear/logistic regression still fall under the umbrella of ML, yet only allow simple relationships to be defined between predictors and outcomes. We recommend that future studies clearly state model assumptions instead of vague statements regarding the use of ML or AI.

Further to the limited reporting of predictive models, many studies lacked clarity in the methodology, making it difficult to understand the process followed for both their model development and implementation phases. Clarification is needed around data processing (i.e. dealing with missing values), model development and validation (i.e. cross-validation, patient numbers) and clinical implementation (i.e. timepoints, outcome measures, success metrics) to confirm the validity of the results. This could be resolved by ensuring that papers adhere to guidelines on trustworthy implementation of AI in healthcare.[42] However, only two articles cited adherence to the International Patient Decision Aid Standards (IPDAS). That being said, within the 11 core domains of the latest IPDAS evidence update, there are no explicit recommendations on AI-incorporated PtDAs.[43] Much like extensions seen with frameworks for prediction model reporting,[44] future IPDAS updates could integrate guidance on AI-driven

PtDAs, as the aims of these tools can align with IPDAS principles of user-centricity to promote SDM.

Prediction methods

Although there was limited reporting regarding model development, those that specified the model often reported using multivariable linear or logistic regression or matching criteria. It is interesting, given the reported performance of nonlinear models in TKA clinical prediction models,[45] that these have not been widely investigated as the underlying model for CDS tool implementations. Studies that reported no significant difference within the implementation phase were often based on linear models.[21,29] Conversely, Twiggs et al.[27] used a nonlinear tree-augmented naïve Bayes model, a more complex ML model, and reported a significant difference in reported pain and change in likelihood for meeting the minimal clinically important difference. There appears to be a link between the flexibility and predictive capacity of the model and implementation success, one that must be considered in the development of models used for decision-making.

Further to this, it is important that the information presented to the patient is both relevant and impactful to the decision-making process. However, there were substantial differences in reported outcome measures between the studies considered in this review. Most studies reported knee pain and physical function as part of the information presented to patients. Indeed, there are suggestions that the high rates of dissatisfaction after TKA could be related to a discordance between patient expectations and the actual level of ongoing pain and reduced function experienced,[46–48] reflecting its importance in patient recovery, and furthermore, at the decision point. Quality-of-life was also reported by some studies. Measures to assess quality-of-life encompass pain, mobility and activities of daily living, individually considered important to patients.[49] However, few studies considered reporting domains beyond this. Some studies report the financial cost of the intervention, and those that did were targeted towards KOA patients earlier on in the journey who were presented with both surgical and non-surgical treatment alternatives.[21,28,30] Further yet, no studies predicted psychosocial outcomes, despite their importance in KOA care.[49]

We included only personalised decision aids that had been applied to a population of patients with KOA, leading to the exclusion of numerous studies that narrowly failed to meet these criteria. This included studies that collected patient information but lacked an underlying prediction model to generate personalised results.[50,51] Emerging proof-of-concept studies demonstrate promising potential,[52–54] coupled with upcoming literature in the field,[55,56] signal a continued shift toward more tailored approaches in KOA CDS tools.

Limitations

Although steps were taken to ensure rigorous database search and analysis of included studies, there were some limitations to this review. First, the limited number of studies included resulted in a largely heterogeneous set of methods and results. The result of this was that only a narrative discussion could be conducted for the review. A second limitation would be the restriction on publication year. Although the search was conducted from 2010 onwards (last 15 years), all but one study has been published within the last 5 years. This reflects the rising trend towards personalisation at the decision point in clinical tools and confirms the validity of the search restriction. However, citation searching uncovered a previous article related to one of the included papers, which was ultimately included in the information in Table 3. Finally, one of the inclusion criteria was that the article should be available in the English language, which could have resulted in articles not included that may have been relevant to the review and potentially increased the geographical distribution of studies undertaken.

Conclusion

Over the next decade, the number of KOA cases is forecast to increase significantly, which will be associated with an escalation in the burden on healthcare services to successfully manage this patient population. As such, there is a growing need to support clinicians to provide effective, targeted and cost-effective treatment, which addresses individual patient needs and optimises SDM practices. AI-based CDS tools offer huge potential to guide KOA treatment decisions. While several CDS tools have been developed to guide KOA treatment strategies, the studies undertaken lack transparency and interpretability in the reporting of ML modelling processes.

Moreover, implementation strategies to date have focused on end-stage KOA treatment discussions or specific patient sub-types, with few developing tools to address treatment strategies for earlier stages of the disease course. These limitations hinder reproducibility and generalisability. Moving forward, there is a need for updated guidelines on AI-driven predictive modelling that can improve reporting standards, strengthen decision aid frameworks and support more robust CDS tool development.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Author contributions

Jodie A. Cochrane: Formal analysis; Investigation; Methodology; Writing – original draft; Writing – review & editing.

Oliver Roberts: Formal analysis; Investigation; Methodology; Writing – original draft; Writing – review & editing.

Karen Ribbons: Investigation; Writing – original draft; Writing – review & editing.

Ross Clark: Conceptualisation; Writing – review & editing.

Yong Hao Pua: Conceptualisation; Writing – review & editing.

Tong Leng Tan: Conceptualisation; Writing – review & editing.

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Ying Ying Leung: Conceptualisation; Writing – review & editing.

Ming Han Lincoln Liow: Conceptualisation; Writing – review & editing.

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Supplemental material

Supplemental material for this article is available online.

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